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# How Model Transformations Affect Fingerprints in Transformer Key–Value Caches

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Model provenance through an inference-time audit channel

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Research
LLM & Agent Security
Model Fingerprinting

Representative work
MISPSO-Attack
Applied Soft Computing 2023

Feature selection
Pattern Recognition 2024

# A released model can lose its visible origin

## Training creates the asset. Distribution makes copies easy.

### Developer

Training + safety
Release terms
Checkpoint W

### Hub / download

Download weights
Copy metadata
Create new versions

### Compliant downstream use

Use or adapt within the release terms

### Suspected unauthorized reuse

Rename or alter weights
Re-host with a different name

**Research need: evidence tied to the model, even when names and metadata change.**

**A fingerprint can support a provenance audit; a copy alone does not prove misuse.**

# A model can also be imitated through its answers

## Example: “Summarize this paragraph” → answer → training pair

### Teacher API

Private model
Receives prompts
Returns answers

→

### Dataset

Input + answer
Many examples
A training set

→

### Training

New parameters
Imitation loss
Learn from data

→

### New service

Similar behavior
A new name
New parameters

**Checkpoint copying transfers weights; distillation transfers behavior through data.**

> **This talk tests transformed checkpoints. Fingerprint inheritance by distilled students remains open.**

# What can establish a model's origin?

| Approach | What it observes | What changes after reuse? |
|---|---|---|
| **Access control** | Who can access an API | Does not follow a downloaded copy |
| **File provenance** | Hash, signature, model card | A modified file has a different hash |
| **Output / trigger tests** | Generated text or a special response | The behavior may change at deployment |
| **Model fingerprint** | An enrolled, verifiable signal | Ask whether the signal survives changes |

> **A fingerprint is an identity signal + a verification procedure. Both must be specified.**

# A fingerprint makes later comparison possible

## Example identity pattern: 1011010 · stored privately before distribution

### Before release

Record the key
Keep a clean model
Embed the ID pattern

### Suspected reuse

Obtain the artifact
Run same documents
Observe internal states

### Verification

Recover the pattern
Measure its strength
Compare to enrollment

Required here: internal model access + matched clean reference.
The result is technical evidence under a stated test, not an automatic ownership verdict.

**The next question is whether deployment preserves enough of this signal to verify it.**

# The KV cache is the model’s working memory

### Ordinary input

Read a document
Predict the next token

### Transformer inference

K: match past tokens
V: reuse their content

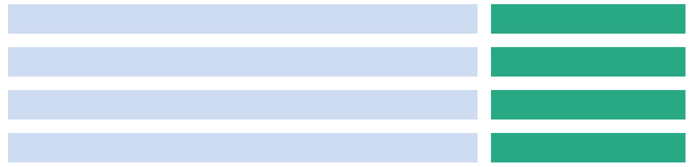
### Next token

Reuse stored K / V
Then the cache grows

## Why compress?

Longer text → more rows
Fewer rows or fewer bits
→ less stored information

## FinCa preserves evidence



Task content  Fingerprint directions

## The conflict

A signal unimportant to text quality can still be important for ownership.

**FinCa protects fingerprint directions during compression, instead of hoping they survive.**

# FinCa recovered the signal on all ten models

| Checkpoint | Exact recovery | Cache ratio |
|---|---|---|
| Qwen3 0.6B | 100% | 3.267× |
| Llama 3.2 1B | 100% | 3.198× |
| Gemma 3 1B IT | 100% | 3.263× |
| Gemma 2 2B | 100% | 3.460× |
| Mistral 7B | 100% | 3.429× |
| Ministral 8B | 100% | 3.220× |
| Llama 3.1 8B | 100% | 3.392× |
| GLM-4 9B | 100% | 3.064× |
| GLM-Z1 9B | 100% | 3.062× |
| DeepSeek V2 Lite | 100% | 3.817× |

## What this established

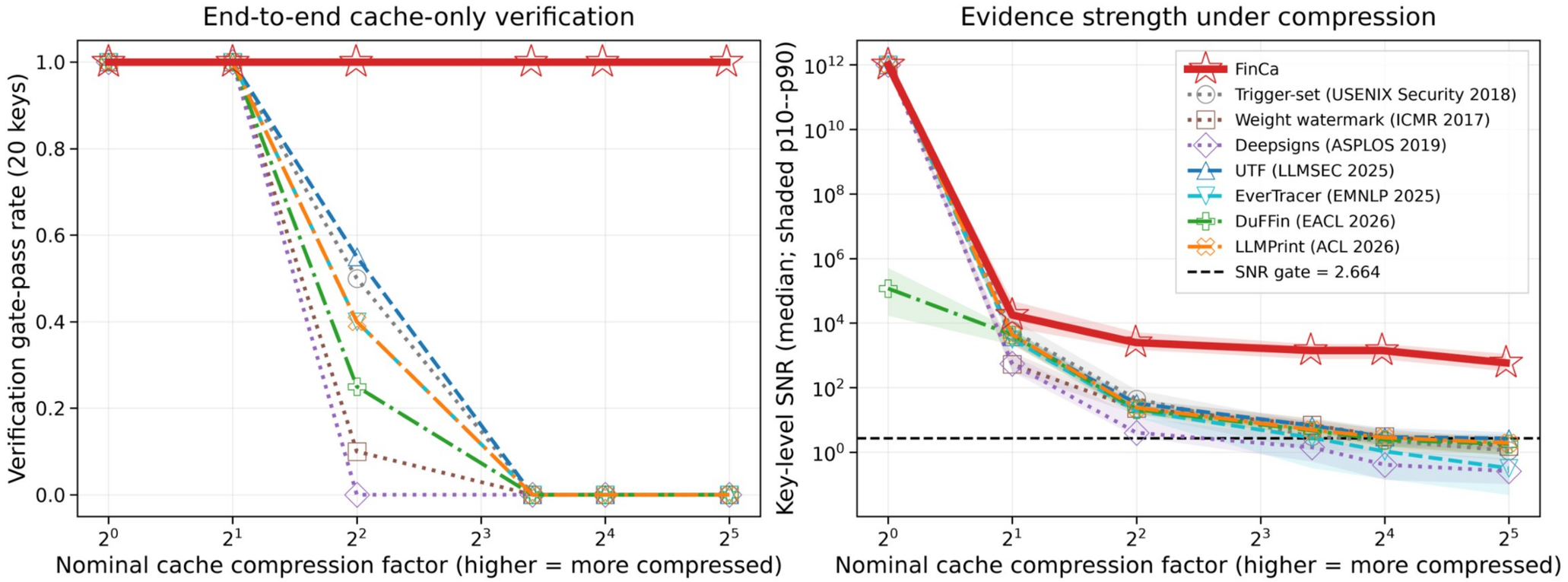
10 **model checkpoints**
20 **held-out documents each**
**Zero bit errors in every row**

## Quality varies by model

Δlog PPL: −0.0385 to +0.1827
Lower is better; 0 = unchanged.
Two physical symbols per model

**Cache ratio = original cache size / complete compressed representation, including decoder state.**

# FinCa preserves evidence as the cache shrinks




Left: verification survives compression. Right: the protected signal stays stronger.

**FinCa protects selected directions during compression. What if later changes do not?**

# FinCa worked. Will the signal last?

## Earlier Table 5: Qwen3-0.6B

**Exact recovery: nominal 2-bit cache / 25% tokens retained***

| | |
|---|---|
| **FinCa** | 100.00% |
| UTF | 11.75% |
| EverTracer | 7.00% |
| DuFFin | 12.75% |
| LLMPrint | 11.00% |

***Two fingerprint directions kept at 6 bits; this is not uniform 2-bit compression.**

## The unresolved question

A downloaded model may later be quantized, mixed, pruned, or run with less cache history.

**Can coding help when these changes do not explicitly protect the fingerprint?**

Adapted baselines under a common cache protocol.

**Present study: measure survival and failure across transformations; do not assume protection.**

# A coded fingerprint after model changes

### Encode

7 bits → 15
coded symbols
Repair some
bit errors

### Interleave

Keyed shuffle
Reorder symbols
Same information
Same energy

### Embed

Value update
Read in V states
Fixed energy
budget

### Decode

Paired V states
Undo permutation
Recover the ID
Measure strength

**Same payload. Same embedding budget.**
**Change one stress condition at a time.**

**Transformation occurs before decoding**

**We ask when redundancy helps, when evidence becomes too weak, and what happens to quality.**

# The two studies answer different questions

| Question | Earlier study: FinCa | Present study |
|---|---|---|
| **Main goal** | Preserve fingerprints during compression by design | Measure survival after controlled transformations |
| **Main lever** | Protect selected directions in a causal cache codec | Add redundancy and compare symbol layouts |
| **Evidence** | Ten-model codec validation with model-specific settings | Four-model matrix with matched coding controls |
| **What we learn** | A protected cache channel can carry audit evidence | Which changes weaken it; when coding helps or fails |

**Feasibility motivates the channel; controlled stress tests reveal its boundaries.**

# Recovery, detection, and quality differ

### Recovery

Did we get the ID?
$1011010 \rightarrow 1011010$
Measure exact recovery
and bit errors

### Acceptance

Is the signal strong?
Correct ID + strength
Measure detection
and false alarms

### Utility

Does the model work?
Compare prediction loss
Measure the change
in log perplexity

## Two different failure patterns to look for

**Correct ID, weak evidence**
**→ decoding succeeds; detection can fail**

**Strong fingerprint, poor predictions**
**→ detection succeeds; quality can fail**

**A reliable conclusion needs the decoded ID, the detection decision, and the quality cost.**

# A controlled test of damage and coding

**4**

**Qwen3 · Qwen2.5**
**Gemma3 · Llama3.2**

**3**

**carrier keys**
**on each checkpoint**

**4**

**BCH layouts**
**Repetition**
**Uncoded control**

**14**

**Precision / mixing**
**Pruning**
**Cache visibility**

## Matched comparison: 7 bits · 15 physical axes · update size 0.04

672 conditions · 26,880 document events
Separate QIM controls: Qwen3 + Llama3.2, one key

**Same budget makes coding comparisons interpretable; the scope remains the tested checkpoints.**

# Redundancy protects a small embedded ID

$\mathbf{b} \in \{0,1\}^7 \rightarrow \mathbf{BCH}(15,7,5) \rightarrow 15$ **signed symbols**

$$\Delta W = \alpha \sum_{i=1}^{15} s_i u_i v_i^T \qquad \|\Delta W\|_F = 0.04$$

## What the code contributes

7 bits identify the message.
15 coded bits add redundancy.
Distance 5: up to 2 hard bit errors.

## What the carrier contributes

$u_i$, $v_i$ select weight directions.
$s_i$ sets each sign; $\alpha$ sets the scale.
The Value states reveal the change.

> **Redundancy is useful only if enough of the embedded signal remains after transformation.**

# Placement changes where damage hits the code

**Interleaving permutes the same 15 symbols; it adds no extra energy.**

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Ordered** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| **Interleaved** | 8 | 2 | 13 | 5 | 10 | 1 | 15 | 7 | 3 | 12 | 6 | 14 | 4 | 11 | 9 |

**Read paired activations → undo permutation → soft decode all 128 messages**

**Accept only if the message matches AND mean signed correlation > 0.5**

> **A correct decoded ID can still be rejected if the measured signal is too weak.**

# One fingerprint, then a concrete model change

## 1 Register an identity

Example ID: 1011010
7 information bits
Key seed: 2027

## 2 Encode and place it

BCH(15,7,5): 15 signs
Interleave over 15 Value axes
Update norm: 0.04

## 3 Mix with clean weights

25% marked + 75% clean
Same model architecture
Fingerprint amplitude shrinks

$$W_{mix} = 0.25\, W_{marked} + 0.75\, W_{clean} = W_{clean} + 0.25\, \Delta W$$

**Measured at mix 0.25: 100% exact recovery, but 0% acceptance.**

Table 4: all four checkpoints, averaged across three keys and 20 marked documents/key.

**Correct bits are insufficient: the signed-correlation score must also exceed 0.5.**

METHOD

# How transformations damage the signal

| Transformation | Controlled intervention | Why it matters |
|---|---|---|
| **Quantization** | Native → INT8 → INT4 → INT2 | Rounding can erase small differences |
| **Weight mixing** | Marked fractions $0.75, 0.50, 0.25, 0$ | The marked update becomes weaker |
| **Projection pruning** | Remove $10\%, 30\%, 50\%$ of coordinates | Carrier information is removed |
| **Cache visibility** | Retain $75\%, 50\%, 25\%$ of token history | The observer has less visible history |

**QIM control: encode a symbol by shifting the quantization grid.**

> **Visibility masking is a controlled intervention; it is not a production cache-eviction benchmark.**

METHOD

# A clean counterpart for every test

### Prepare data

12 selection docs
16 calibration docs
20 held-out pairs

### Run paired routes

Clean + marked model
Same documents
Same transformation

### Report outcomes

Detection / false alarms
Recovery / bit errors
Prediction quality

## Matched controls for coding

BCH interleaved · BCH ordered · interleaved repetition · uncoded
Same 7-bit payload, 15 axes, and perturbation norm in every construction.

**Document events are the statistical unit; changing the key does not create a new model population.**

# How an ordinary document becomes an audit

**Prompt · WikiText-2 test document**

The Ise @-@ class ships'waterline protective belt had a maximum thickness of 299 mm ( 11

**C0 · Clean**

.7 in ) and a minimum thickness of 200 mm ( 7.9 in ). The belt was made of 0.5 mm ( 0.02 in ) thick steel and was attached to the ship's hull by 1.5 mm ( 0.06 in ) thick steel rivets

**F0 · Fingerprinted**

.7 in ) and a minimum thickness of 200 mm ( 7.9 in ). The belt was made of 0.5 mm ( 0.02 in ) thick steel and was attached to the ship's hull by 1.5 mm ( 0.06 in ) thick steel rivets

**C1 · Clean + codec**

.7 in ) and a minimum thickness of 150 mm ( 5.9 in ). The belt was made of 0.5 mm ( 0.020 in ) thick steel and was attached to the ship's hull by 12 bolts. The belt was designed to protect the ship's hull from damage

**F1 · Fingerprinted + codec**

.7 in ) and a minimum thickness of 100 mm ( 3.9 in ). The belt was made of 0.5 mm ( 0.02 in ) thick steel and was attached to the ship's hull by 12 rivets. The belt was designed to protect the ship's hull from

FinCa example: text changes are context, not the audit rule.

## 1 Use the same document

Clean + marked routes
Same transformation

## 2 Compare Value states

Marked minus clean
Read 15 carrier correlations

## 3 Decode and decide

Correct ID AND score > 0.5
Count 20 paired test docs/key

**Present study: WikiText-2 validation material; 12 selection / 16 calibration / 20 test pairs.**

# Detection depends on the transformation

| | Qwen3 | Qwen2.5 | Gemma3 | Llama3.2 |
|---|---|---|---|---|
| **Native** | 100.0 | 100.0 | 100.0 | 100.0 |
| **INT8** | 100.0 | 100.0 | 100.0 | 100.0 |
| **INT4** | 100.0 | 100.0 | 76.7 | 100.0 |
| **INT2** | 5.0 | 0.0 | 0.0 | 0.0 |
| **Mix** 0.75 | 100.0 | 100.0 | 100.0 | 100.0 |
| **Mix** 0.50 | 61.7 | 28.3 | 30.0 | 28.3 |
| **Mix** 0.25 | 0.0 | 0.0 | 0.0 | 0.0 |
| **Mix** 0 | 0.0 | 0.0 | 0.0 | 0.0 |
| **Prune** 10% | 100.0 | 100.0 | 100.0 | 100.0 |
| **Prune** 30% | 78.3 | 100.0 | 83.3 | 100.0 |
| **Prune** 50% | 43.3 | 93.3 | 8.3 | 93.3 |
| **Keep** 75% | 100.0 | 100.0 | 100.0 | 100.0 |
| **Keep** 50% | 100.0 | 100.0 | 100.0 | 100.0 |
| **Keep** 25% | 100.0 | 100.0 | 100.0 | 100.0 |

**Native / INT8**
100% **TPR**

**INT2**
0–5% **TPR**

50% **pruning**
8.3–93.3%

Cells = acceptance rate (%) · 60 marked events/model · Higher is better · Table 1 / Figure 1

# Recovery vs Acceptance vs Utility

| | Recovery | Acceptance | Utility |
|---|---|---|---|
| **Mix** 0.25<br>25% **marked weight** | 100%<br>exact message | 0%<br>thresholded TPR | ≈0<br>Δlog PPL |
| **Keep** 25%<br>**visibility** | 100%<br>exact message | 100%<br>thresholded TPR | +4.90–5.96<br>Δlog PPL |

Recovery ≠ detection ≠ model quality · Δlog PPL: lower is better · All four checkpoints · Table 4

# Coding helps, but it does not win in every setting

## Alternative carrier: QIM

50% pruning · BCH / uncoded detection · offset 0

| | |
|---|---|
| **Qwen3** | 55% / 0% |
| **Llama3.2** | 100% / 0% |

## No universal winner

Gemma3, 50% pruning
Repetition: 15.0% TPR
BCH interleaved: 8.3%

Gains depend on the model, carrier and transformation

QIM: Tables 2–3, one key / two models · Repetition vs BCH: Table 5 · TPR = detection rate

NEXT STEPS

# From a protected channel to evidence that lasts

| | | |
|---|---|---|
| 01 | **Active attackers** | Deliberately target carrier directions; measure removal cost. |
| 02 | **Broader evidence** | All-family QIM; more checkpoints, layers, keys, and task metrics. |
| 03 | **Real deployment** | Cache-management policies, longer histories, and adaptation. |
| 04 | **New lineage routes** | Test whether any fingerprint evidence survives distillation. |

> **FinCa: protect evidence by design. Present study: measure where that evidence survives or fails.**